

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning **(BL 5)**

Assessment Tool Weightage: 10%

QUESTIONS: 15

Total Marks:20

Annexure A

Comparative Analysis

Code of Conduct:

*I **P.Sai Ganesh Reddy(192525111)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P.Sai Ganesh Reddy

Annexure A

Short Answer Test

1. Compare K-Nearest Neighbour and Case-Based Learning using a real-world example such as medical diagnosis. Analyze differences in memory usage, similarity measures, and prediction strategy.
2. Compare Locally Weighted Regression and Radial Basis Function networks for function approximation tasks. Use an example dataset to explain differences in locality, computational cost, and smoothness of predictions.
3. Analyze the differences between K-Nearest Neighbour and Radial Basis Function models in classification problems. Provide an example and compare decision boundaries, training time, and generalization ability.
4. Compare Case-Based Learning and Locally Weighted Regression using a practical example such as recommendation systems. Discuss how each approach handles new data, adaptability, and computational efficiency.
5. Compare Decision Tree Learning and K-Nearest Neighbour (KNN) using a classification problem such as customer churn prediction. Analyze differences in model interpretability, training time, handling of noisy data, and decision-making process. Discuss how each method performs with large datasets and complex feature spaces.
6. Compare Naïve Bayes and Logistic Regression for a text classification task such as spam detection. Analyze assumptions about feature independence, probability estimation, computational efficiency, and performance on high-dimensional data. Discuss scenarios where one method outperforms the other.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	15	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	20	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1.Compare K-Nearest Neighbour and Case-Based Learning using a real-world example such as medical diagnosis. Analyze differences in memory usage, similarity measures, and prediction strategy.

ANSWER:

Example: Medical Diagnosis

Suppose a patient has symptoms such as fever, cough, fatigue, and chest pain. A system can compare the patient's symptoms with previous patient cases to predict a possible disease.

Feature	K-Nearest Neighbour (KNN)	Case-Based Learning (CBL)
Basic idea	Finds the K most similar training samples	Retrieves and uses similar previous cases
Memory usage	Stores most/all training data	Stores cases in a case library
Similarity measure	Usually Euclidean, Manhattan, cosine, etc.	Domain-specific similarity measures can be used
Prediction	Majority vote for classification or average for regression	Adapts/reuses the solution of similar cases
Learning	Very little explicit training	Learns by retaining and reusing past cases
Adaptability	New examples can be added easily	New cases and solutions can be added to the case base
Example	Find 5 patients with similar symptoms and use their diagnoses	Retrieve a previous patient with similar symptoms and adapt the diagnosis

Conclusion:

KNN is mainly a distance-based prediction technique, whereas Case-Based Learning emphasizes **retrieving, reusing, revising, and retaining previous cases**. CBL can use richer domain-specific similarity and adaptation strategies.

2. Compare Locally Weighted Regression and Radial Basis Function networks for function approximation tasks. Use an example dataset to explain differences in locality, computational cost, and smoothness of predictions.

ANSWER:

Example Dataset:

Consider a dataset containing house size and house price:

House Size (sq.ft.) Price (₹ lakhs)

House Size (sq.ft.) Price (₹ lakhs)

800	35
1000	42
1200	50
1500	65
1800	78
2200	95

Comparison

Feature	Locally Weighted Regression	RBF Network
Locality	Fits a model around each query point	Uses localized radial basis functions
Training	Little/no global training	Requires learning centers, widths and output weights
Prediction cost	High because nearby points must be considered for every query	Usually faster after training
Memory	Requires training data at prediction time	Stores learned parameters
Smoothness	Depends strongly on bandwidth/kernel	Usually produces smooth continuous predictions
Adaptability	Very high to local data changes	Requires retraining/update of network
Suitable for	Small datasets and locally varying relationships	Larger function-approximation problems

In locally weighted regression, nearby house-size observations receive **higher weights**, while distant observations receive lower weights.

An RBF network instead uses functions such as:
where μ is the centre and controls the spread.

Conclusion:

Locally Weighted Regression is highly local and flexible but can be computationally expensive during prediction. RBF networks learn localized basis functions and generally provide smoother and faster predictions after training.

3. Analyze the differences between K-Nearest Neighbour and Radial Basis Function models in classification problems. Provide an example and compare decision boundaries, training time, and generalization ability.

ANSWER:

Example: Classification of Fruits

Suppose fruits are classified as **Apple** or **Orange** using weight and diameter.

KNN

For a new fruit, KNN calculates its distance from training examples and selects the nearest samples.

For example, with :

- 4 nearest fruits → Apple
- 1 nearest fruit → Orange

Therefore, the new fruit is classified as **Apple**.

RBF Model

An RBF network has hidden neurons with radial basis functions. Each neuron responds strongly to inputs close to its centre. The outputs of these neurons are combined to determine the class.

Feature	KNN	RBF
Training time	Very low	Higher
Prediction	Can be computationally expensive	Generally faster after training
Decision boundary	Can be irregular and highly local	Generally smoother
Memory	Stores training examples	Stores learned model parameters
Generalization	Sensitive to , distance and noise	Can generalize smoothly depending on centres/widths
Updating data	Easy	May require retraining

Decision Boundary:

KNN can create complex boundaries because decisions depend directly on nearby samples. RBF models create smoother boundaries based on combinations of radial basis functions.

Conclusion:

KNN is simple and effective for smaller datasets, while RBF models are useful when a learned nonlinear model and smoother generalization are required.

4. Compare Case-Based Learning and Locally Weighted Regression using a practical example such as recommendation systems. Discuss how each approach handles new data, adaptability, and computational efficiency.

ANSWER:

Example: Recommendation System

Consider a movie recommendation system. A user has watched several movies and rated them. The system needs to recommend a new movie.

Case-Based Learning

CBL searches its case library for users/cases similar to the current user.

Process:

1. Receive the new user's preferences.
2. Retrieve similar previous cases.
3. Examine their successful recommendations.
4. Adapt the recommendations to the new user.
5. Store the new experience as a future case.

Locally Weighted Regression

LWR uses nearby observations to estimate a continuous value such as the predicted rating of a movie.

Nearby users/movies receive higher weights, while distant observations receive lower weights.

Feature	Case-Based Learning	Locally Weighted Regression
New data	Retrieves similar previous cases	Uses nearby data points
Adaptability	Very high through case reuse and adaptation	High because local models are fitted
Output	Usually a reused/adapted solution	Numerical prediction
Computational efficiency	Retrieval can be costly for a large case base	Prediction can be costly because local fitting is repeated
Knowledge representation	Cases	Numerical training data
Best suited for	Problems with reusable experiences	Continuous prediction

Conclusion:

CBL is preferable when previous experiences and solutions are important, whereas LWR is more suitable for predicting numerical outcomes from locally related data.

5. Compare Decision Tree Learning and K-Nearest Neighbour (KNN) using a classification problem such as customer churn prediction. Analyze differences in model interpretability, training time, handling of noisy data, and decision-making process. Discuss how each method performs with large datasets and complex feature spaces.

ANSWER:

Example: Customer Churn Prediction

A company wants to predict whether a customer will **churn** or **stay** using features such as:

- Age
- Monthly charges
- Contract type
- Number of complaints
- Tenure

Comparison

Feature	Decision Tree	KNN
Training time	Usually higher than KNN	Very low
Prediction time	Fast	Can be slower
Interpretability	High; rules can be easily understood	Low
Noisy data	Can overfit; pruning helps	Sensitive to noisy/irrelevant neighbours
Decision process	Uses hierarchical if-then rules	Uses similarity to nearest samples
Large datasets	Training can be expensive, but prediction is efficient	Prediction becomes expensive
Complex feature spaces	Can handle nonlinear relationships	Can work well but suffers in high dimensions
Feature scaling	Generally not required	Usually important
Memory	Stores learned tree	Stores training examples

Example Decision Tree Rule

A tree might produce:

If contract = month-to-month

→ **and** monthly charges are high

→ then predict **Churn**.

KNN instead finds customers with similar characteristics and uses their majority class.

Conclusion:

Decision Trees are preferable when interpretability and fast prediction are important. KNN is attractive when training simplicity and easy incorporation of new training examples are more important. For very large or high-dimensional datasets, KNN can become computationally expensive and less effective because of distance-related problems.

6. Compare Naïve Bayes and Logistic Regression for a text classification task such as spam detection. Analyze assumptions about feature independence, probability estimation, computational efficiency, and performance on high-dimensional data. Discuss scenarios where one method outperforms the other.

ANSWER:

Example: Spam Email Detection

The objective is to classify an email as:

- **Spam**
- **Not Spam**

Features may include words such as *free*, *offer*, *winner*, *meeting*, and *project*.

Naïve Bayes

Naïve Bayes uses Bayes' theorem:

It assumes that features are conditionally independent given the class.

For example:

is calculated using the probabilities of the observed words given that the email is spam.

Logistic Regression

Logistic Regression estimates the probability of a class using:

It learns weights for the input features and uses the resulting probability for classification.

Feature	Naïve Bayes	Logistic Regression
Main assumption	Features are conditionally independent	No feature-independence assumption
Probability estimation	Based on feature/class probabilities	Learns model weights
Training	Very fast	Usually requires iterative

Feature	Naïve Bayes	Logistic Regression
		optimization
High-dimensional data	Excellent, especially sparse text	Also very effective
Data requirement	Can perform well with relatively small datasets	Often benefits from more training data
Feature relationships	Independence assumption may be unrealistic	Can model correlated features better
Interpretability	Probabilistic	Feature weights can be interpreted

When Naïve Bayes Performs Better

Naïve Bayes is often preferable when:

- The dataset is very large and sparse.
- Fast training is required.
- There are many text features.
- A simple probabilistic model is sufficient.

When Logistic Regression Performs Better

Logistic Regression can outperform Naïve Bayes when:

- Features are correlated.
- The independence assumption of Naïve Bayes is unsuitable.
- A discriminative decision boundary is preferred.
- Properly regularized linear classification is appropriate.

Conclusion:

For text-based spam detection, both methods are strong choices. Naïve Bayes is particularly attractive for its **simplicity and computational efficiency**, while Logistic Regression can provide better performance when feature dependencies and a discriminative decision boundary are important.